Does AI Improve Customer Satisfaction?

An Analysis of Customer Complaints to US Banks

Group 10

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AGENDA

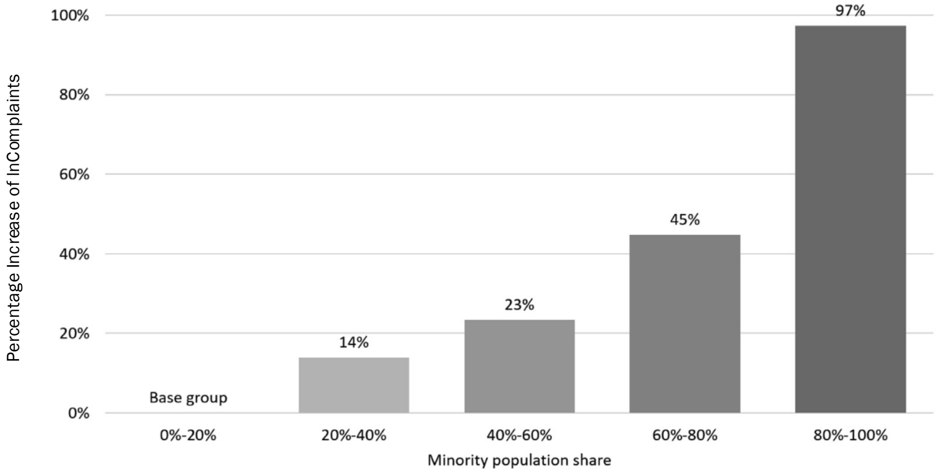
1. Previous Literature
2. Methodology & Dataset Description
3. Differences between AI and non-AI Banks Complaints
4. Impact of AI Adoption by Banks
5. Conclusion
6. Limitations & Future Researches

PREVIOUS LITERATURE

Begley et al. (2021) - *Quantitative*

Low-income and minority neighborhoods receive **significantly lower-quality** financial services, evidenced by **much higher CFPB complaint rates**

Regulatory pressure from the CRA to **increase lending volume** further raises complaints, revealing a **quantity-quality trade-off** that **disproportionately harms minority communities**



Source: Begley, T. A., & Purnanandam, A. (2021). Color and credit: Race, regulation, and the quality of financial services. Journal of Financial Economics, 141(1), 48-65.

Graham et al. (2025) - *Qualitative*

Complaint Reduction Factor

- \+ Provide **instant, 24/7 responses**, reducing wait-time frustration
- \+ **Handle simple, routine queries effectively** (e.g., FAQs, basic information)
- \+ **Offer convenience and consistent information**, improving customer experience for straightforward tasks
- \+ **Improve operational efficiency**, reducing internal workload and speeding up service delivery

Complaint Reduction Factor

- \- **Struggle with complex or multi-step issues**, leading to unresolved problems
- \- **Lack emotional capability**, making them unsuitable for sensitive or high-stakes interactions
- \- **Frustrate users when they cannot easily reach a human**, especially when the chatbot fails
- \- **Create mistrust due to security and privacy concerns**

Source: Graham, G., Nisar, T. M., Prabhakar, G., Meriton, R., & Malik, S. (2025). Chatbots in customer service within banking and finance: Do chatbots herald the start of an AI revolution in the corporate world?. Computers in Human Behavior, 165, 108570.

METHODOLOGY

1. Loading the Data

Imported data for 2015-2025 (**1.5M+ complaints**) from the cfpb database

cfpb Consumer Financial Protection Bureau

+

Combined it with **demographical data** at the ZIP code level

United States Census Bureau

2. Filtering & Cleaning

Filter on 2015-2023 due to **unavailability of data on income** for more recent years

We excluded institutions with **fewer than 260 complaints** (the overall average) leaving **233 financial institutions**

Removed complaints with **incomplete zip codes**

= ~440k complaints

3. AI Labelling

Used **ChatGPT** and **Gemini** to determine whether each of the 233 institutions had a **customer-service chatbot** and, if so, **when it was introduced**

Manually resolved LLM disagreements
(assigned "Unknown" when no date could be verified and used Jan 1 when only the year was known)

= 53 AI chatbots adopter
(63k complaints AI, 30k pre-AI, 8k Unknown)

4. Sentiment Analysis

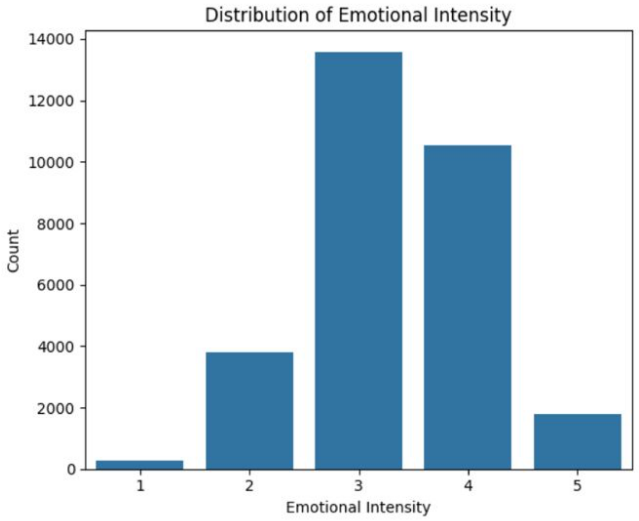
Sampled **30k complaints** and used the GPT API to score **emotional intensity**, **tone**, and **harm severity** using a **standardized labeling prompt**

Trained a **DeBERTa model** on 27k labeled complaints and validated it on the remaining 3k

Applied the DeBERTa model to the **remaining 410k complaints**

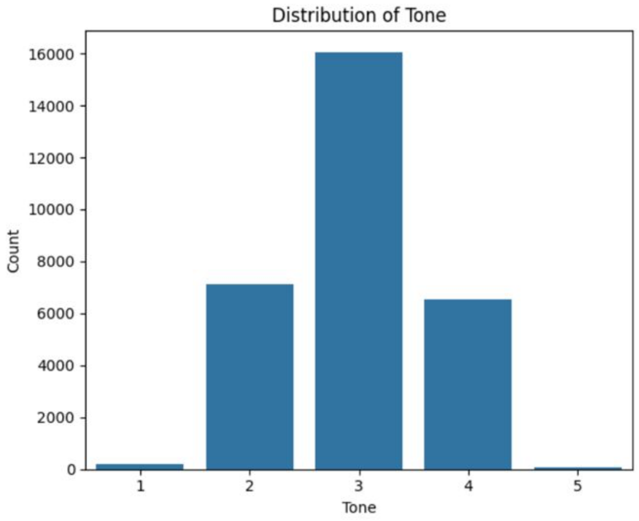
A LOOK AT THE DISTRIBUTION OF OUR DATASET

Distribution of Emotional Intensity



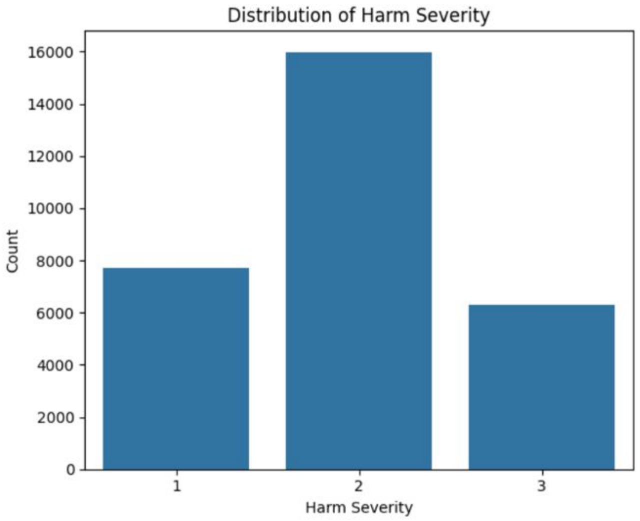
1 = mild annoyance
2 = dissatisfied but calm
3 = clearly angry or upset
4 = very angry, strong language or repeated failures
5 = extreme outrage or life-impacting harm

Distribution of Tone



1 = polite
2 = neutral
3 = frustrated
4 = hostile
5 = threatening

Distribution of Harm Severity

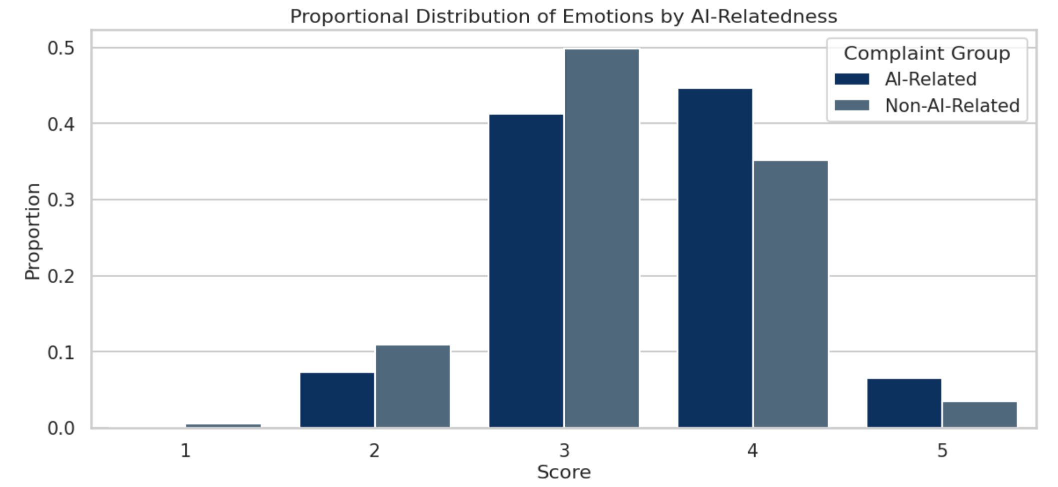


1 = none_low
2 = medium
3 = high

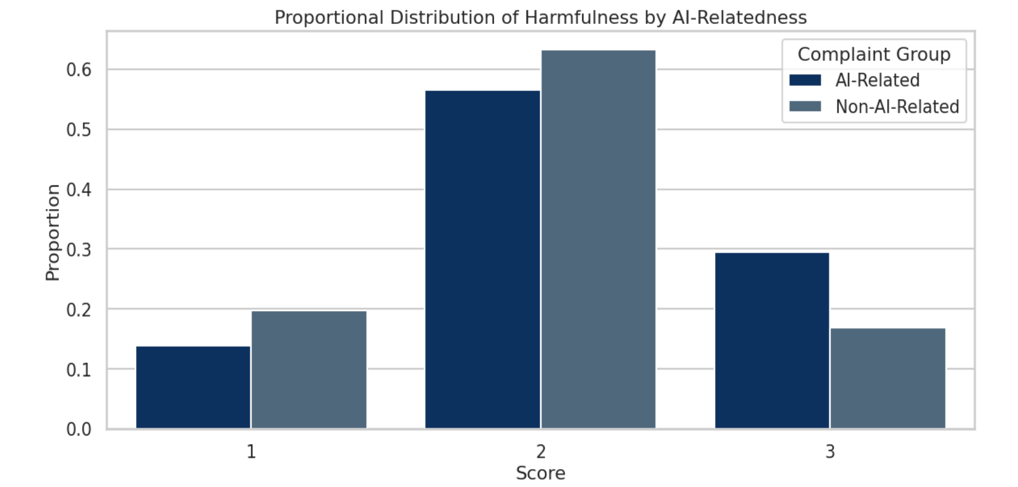
Bocconi

SENTIMENT ANALYSIS BETWEEN AI AND NON-AI BANKS

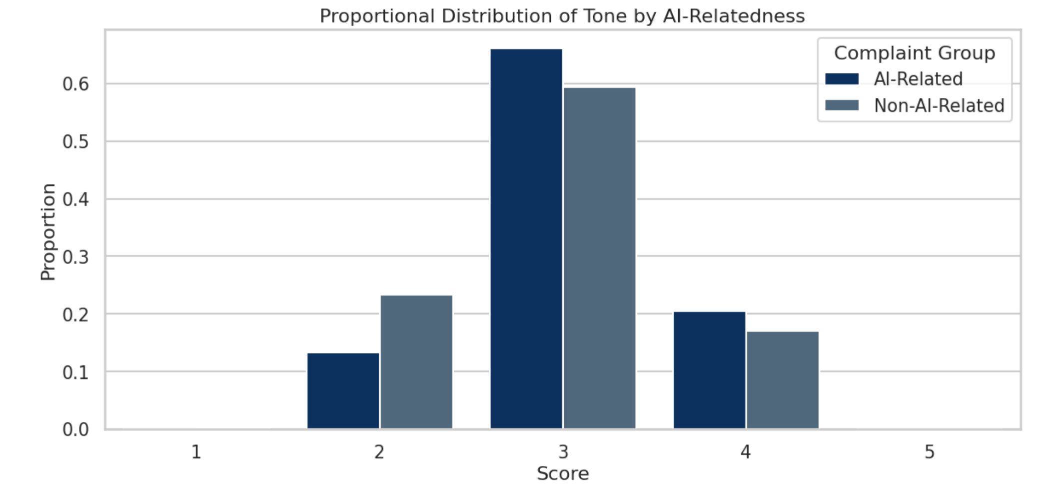
Emotions



Harmfulness



Tone

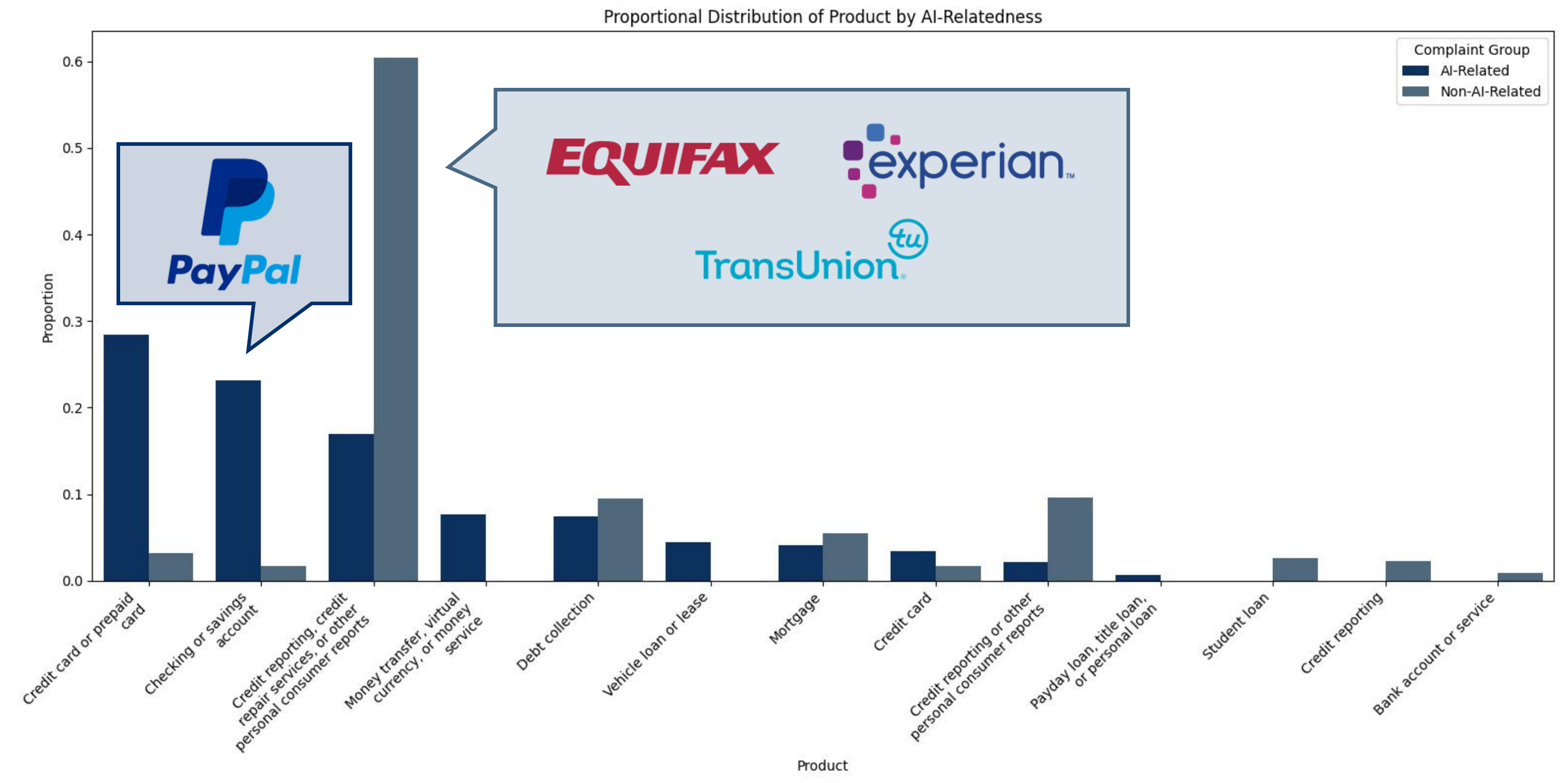


	AI-Related	Non-AI-Related
	Ø = 3.50	Ø = 3.30
	Ø = 3.07	Ø = 2.93
	Ø = 2.26	Ø = 1.97

COMPLAINTS DISTRIBUTION PER PRODUCT

Products

Large differences in the **distribution of products between AI and non-AI banks, largely** driven by the "**Big Three" US Credit Bureau** (Experian, Equifax & TransUnion), but also by **difference in business models** of remaining banks



IS THIS AN AI-RELATED OR A BANK-SPECIFIC CONCLUSIONS?

Simply comparing banks that adopted AI with those that did not can **distort the true effects** of AI adoption and lead to **incorrect inferences:**

1. AI adopters may be systematically different to begin with:

Banks facing the **highest volume or most urgent types** of customer complaints may be **more likely to adopt AI tools,** meaning adoption is driven by **pre-existing pressure rather than AI effectiveness.**

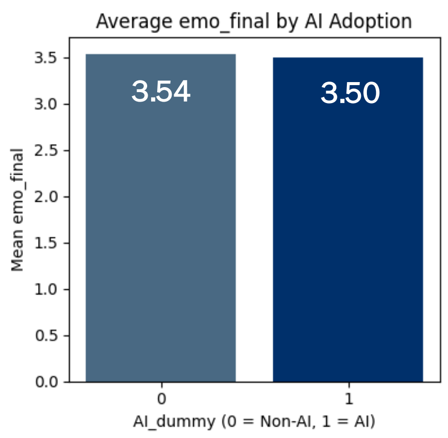
2. Differences in complaint patterns may reflect business models, not AI:

Banks receive **different kinds of customer inquiries** depending on their size, product mix, and customer base. Apparent differences in complaint levels might therefore **stem from underlying business characteristics, not from AI deployment.**

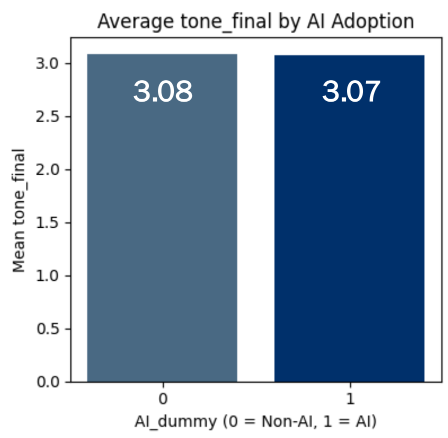
DOES AI ADOPTION INCREASE CUSTOMERS’ SENTIMENT

When looking at overall complaints' evolution following AI adoption **no clear sentiment improvement** is visible

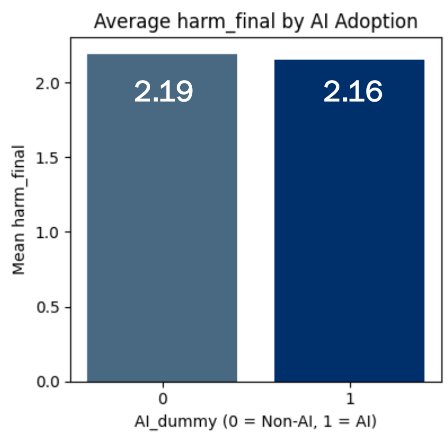
Emotions



Tone

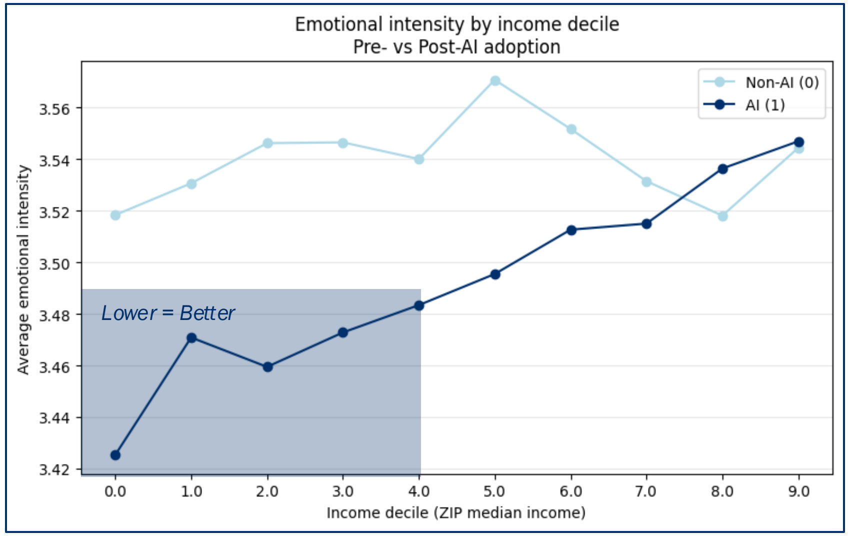


Harmfulness



We need to look further through the US Census Bureau’s data

HETEROGENEITY ACROSS INCOME DISTRIBUTION



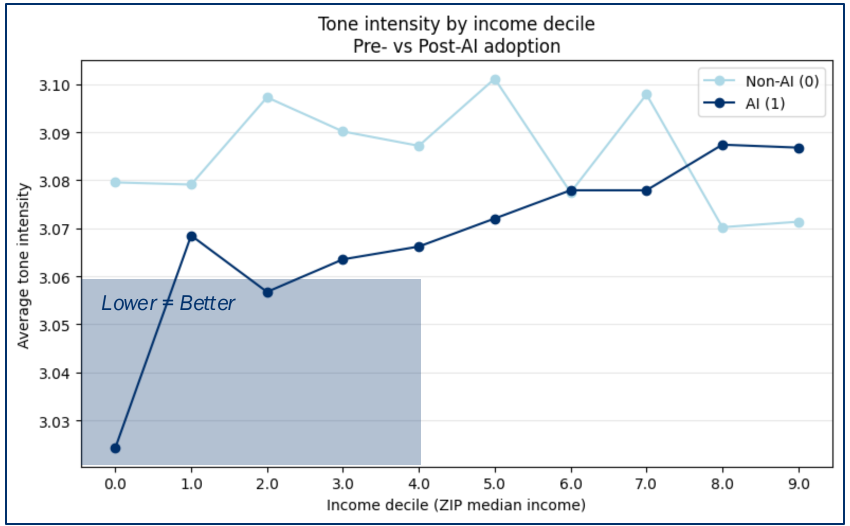


Table 1: Income heterogeneity in the effect of AI on sentiment dimensions

	Emotional intensity (1)	Tone (2)	Harm severity (3)
AI (income decile 0)	-0.036	-0.037**	-0.024
	(0.023)	(0.016)	(0.020)
AI × income decile 7	0.065*	0.034	0.050*
	(0.030)	(0.025)	(0.026)
AI × income decile 8	0.098***	0.071***	0.055*
	(0.023)	(0.021)	(0.022)
AI × income decile 9	0.079*	0.069***	0.064**
	(0.038)	(0.026)	(0.030)
Observations		93,992	
Fixed effects		Bank FE, Year–Month FE	
Standard errors		Clustered at bank level	

Notes: Each column reports a separate regression. Dependent variables measure different dimensions of complaint sentiment: emotional intensity (1–5), tone (1–5), and harm severity (1–3). All models include bank fixed effects and year–month fixed effects. Standard errors are clustered at the bank level. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

When comparing **customer complaint rates** across AI-adopting and non-adopting banks, the most substantial differences appears in the **lower income deciles**, where AI adoption is associated with complaints with better sentiment. However, in the **top income decile**, complaint sentiment increases, suggesting that AI has a **negative impact** to higher-income consumers.

HETEROGENEITY ACROSS RACIAL COMPOSITION

Table 2: AI Adoption and Racial Diversity Heterogeneity

	Emotional intensity	Tone	Harm severity
	(1)	(2)	(3)
AI dummy	0.019	0.023*	0.013
	(0.017)	(0.014)	(0.017)
Diversity score ^	-0.006	0.0089	0.008
	(0.022)	(0.013)	(0.018)
AI dummy × Diversity score	-0.0483*	-0.056***	-0.027
	(0.028)	(0.018)	(0.028)
Observations		93,893	
Fixed effects		Bank FE, Year–Month FE	
Standard errors		Clustered at bank level	

Notes: See Notes in Table 1.

^Diversity score = 1 - % of other races

Insights

- The effect of AI chatbot adoption **varies with district racial diversity**
- In **more diverse ZIP codes**, AI adoption is associated with a **relative softening of complaint sentiment**, reflected in a negative interaction across sentiment measures
- The impact is negative, small and marginally significant for Emotional Intensity (10%) and highly significant for tone (1%).
- The **main effects of AI and diversity are near zero**, indicating **no overall sentiment shifts and no baseline differences across ZIP codes**

AI's influence on complaint sentiment is **not uniform across communities,** motivating **deeper analysis of group-specific racial heterogeneity**

HETEROGENEITY ACROSS RACIAL COMPOSITION

Table 3: AI Effects Across Racial Groups

	Emotional intensity (1)	Tone (2)	Harm severity (3)
AI dummy	0.020	0.017	0.015
	(0.016)	(0.014)	(0.015)
% Black	-0.039	-0.021	0.009
	(0.032)	(0.023)	(0.040)
% Hispanic	0.029	0.018	0.061
	(0.034)	(0.020)	(0.037)
% Asian	-0.105***	-0.071**	-0.118***
	(0.033)	(0.031)	(0.036)
AI dummy × % Black	-0.136***	-0.096***	-0.108***
	(0.028)	(0.020)	(0.032)
AI dummy × % Hispanic	-0.033	-0.026	-0.030
	(0.035)	(0.017)	(0.033)
AI dummy × % Asian	0.082	0.029	0.120**
	(0.050)	(0.033)	(0.054)

Notes: See Notes in Table 1.

Insights

- **No overall shift in sentiment** following AI adoption
- ZIP codes with **higher Black population shares** show substantially **larger decreases in sentiment scores**, indicating less negative complaint sentiment after adoption
- Some weak evidence of **increased harm severity** in ZIP codes with **higher Asian population shares**, though this pattern is **not consistent across sentiment dimensions**

*Note: These results hold even **when controlling for income**, suggesting that the observed racial patterns are **not simply driven by socioeconomic differences***

CONCLUSIONS

1 AI adoption does not change overall complaint sentiment

Across banks, sentiment remains broadly unchanged before and after AI implementation.

2 AI adoption worsens sentiment in high-income ZIP codes

While AI is associated with modest sentiment improvements in lower-income areas, the pattern reverses at the top of the income distribution, where AI adoption leads to **more negative or hostile sentiment.**

3 Racial disparities are limited

AI's impact **varies little across racial compositions**, with only minor tone improvements in more diverse ZIP codes

The implementation of AI **does** not appear to meaningfully shift sentiment **(neither positively nor negatively) on average**, with effects emerging only **in specific income segments**

LIMITATION & FUTURE RESEARCHES

1

Unobserved complaint channel

We cannot determine whether a **complaint was generated through the chatbot** or via **another customer-service route**, making it difficult to isolate chatbot-specific effects

2

Income measured at the area level, not the customer level

ZIP-code median income is only a proxy, we do not observe the **actual income of individual complainants**, which may introduce measurement noise

3

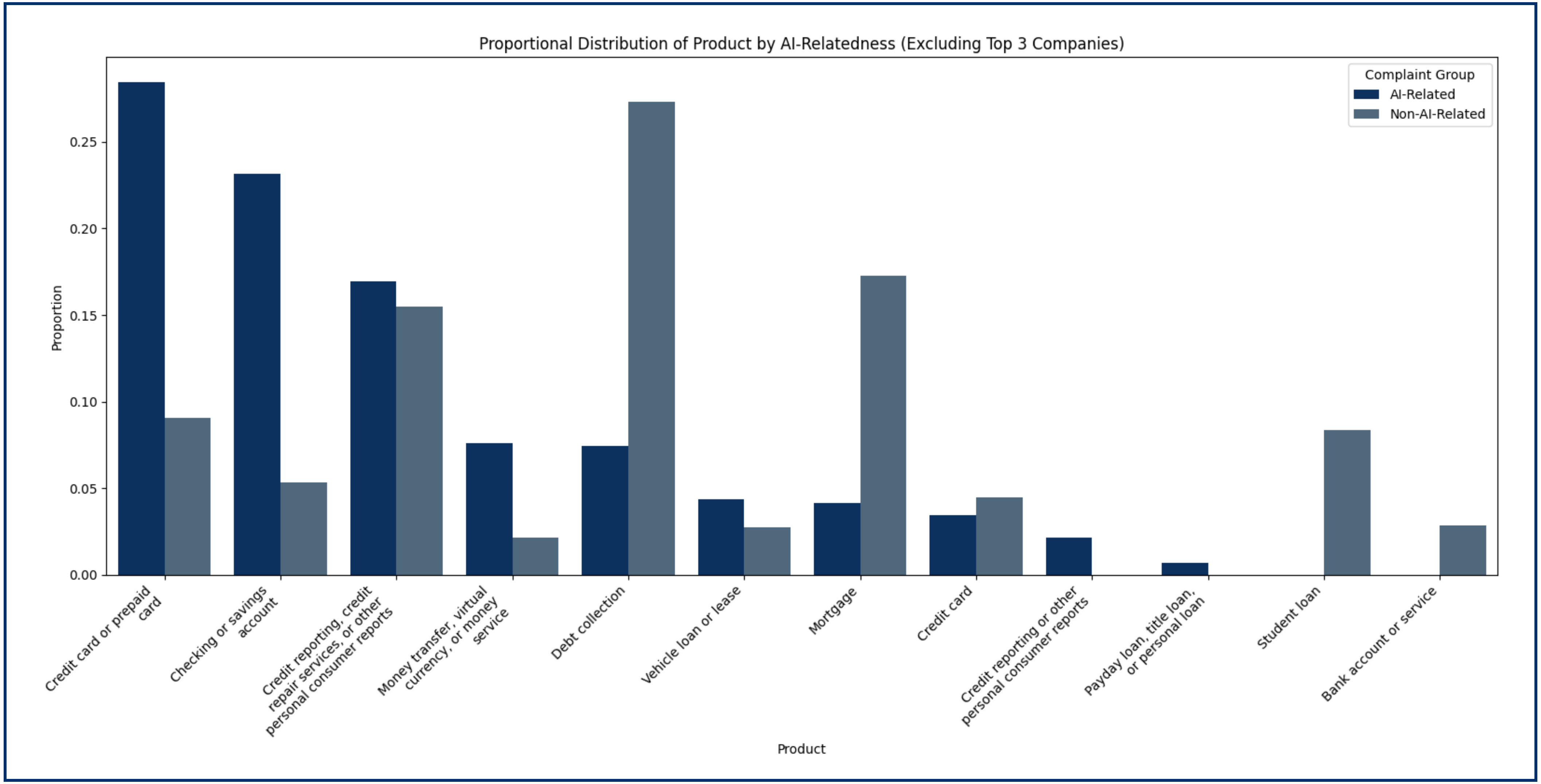
Incomplete information on AI implementation timing

The exact rollout dates of chatbot adoption **are not always known**, limiting the precision of before-and-after comparisons (use of "Unknown" / Jan. 1st)

Future research would require **richer datasets** and should aim to link complaints to **specific chatbot interactions**, incorporate **individual-level income information**, and use **precise implementation timelines** to more accurately identify the true impact of AI on customer sentiment


Thank You!

APPENDIX – DISTRIBUTION BY PRODUCT (EXCLUDING BIG THREE)



APPENDIX – RACIAL AND INCOME COMPOSITION

Table 4: Income and Racial Moderators of AI Adoption Effects

	Emotional intensity (1)	Tone (2)	Harm severity (3)
AI dummy	-0.022 (0.028)	-0.008 (0.021)	0.000 (0.029)
Median income	-0.00000021 (0.00000022)	-0.00000029** (0.00000014)	0.00000001 (0.00000017)
AI dummy × Median income	0.00000049* (0.00000026)	0.00000030 (0.00000019)	0.00000017 (0.00000024)
% Black	-0.049* (0.028)	-0.034 (0.023)	0.001 (0.041)
AI dummy × % Black	-0.111*** (0.030)	-0.082*** (0.019)	-0.099*** (0.036)
% Hispanic	0.019 (0.031)	0.004 (0.020)	0.062* (0.037)
AI dummy × % Hispanic	-0.009 (0.035)	-0.011 (0.020)	-0.021 (0.036)
% Asian	-0.086*** (0.031)	-0.046 (0.028)	-0.118*** (0.042)
AI dummy × % Asian	0.035 (0.043)	0.004 (0.029)	0.103* (0.059)

Notes: See Notes at Table 1.

APPENDIX – PRODUCT SHARE WITHIN INCOME DISTRIBUTION AND % BLACK